

Founding ML Scientist specializing in AI, LLM, agent, and unsupervised graph models. 20 patents. 5 enterprise customers across the US, India, and the Gulf.

Work Experiences

- 2024 – Present Founding ML Scientist, RaptorX.AI, Remote
- Built a production-style Agentic Engineering workflow for bank fraud alerts: Agent Loop, context management, planning, Guardrails, case-store state, Evidence/Report agents, SFT/RLVR/RLHF, evaluation, and runtime/MLOps
 - Built criteria-driven multi-agent systems for paper/patent production: literature/gap scans, experiment checks, TMLR gates, prior-art search, drafting, counsel handoff, and audit logs
 - Built the underlying unsupervised fraud engine from zero: GNNs, behavioral autoencoders, explainability, and a 30-type edge-probability model to reduce label dependency
 - Achieved $328\times$ candidate reduction ($590K \rightarrow 1,883$ high-confidence alerts), 99.9% fraud-type specificity
 - Converted PoCs into 5 signed enterprise customers across the US, India, and the Gulf: Payactiv (US), NSDL and Lenskart (India), Omantel and Thawani Pay (Oman); US deployment covers 460K+ users across 6,700+ employers
- 2021 – 2024 Senior Engineer, AI & Data Platform for Autonomous Driving, Lotus Tech (Geely Group), Hangzhou, China
- Led a fully in-house video anonymization pipeline from data-collection vehicles (80+ TB/day) to data center. YOLOv5-based model with 90%+ recall on faces and license plates across 10M+ frames, ensuring GDPR/PIPL compliance
 - Reduced latency 30% via GPU-accelerated inference at fleet scale; saved ~\$1M/year by replacing third-party vendors entirely
- 2020 – 2021 Independent ML Study & Graph-ML Projects
- Self-directed study of graph machine learning: implemented GNN models for node classification and link prediction on public Kaggle datasets
- 2020 Software Development Engineer, Financial Anti-Fraud, DataVisor, Shanghai, China
- Production Fraud Detection for Top-Tier Commercial Bank:
 - Built an unsupervised loan application fraud model combining GNN and traditional ML, using hundreds of features spanning behavioral, device, third-party credit, and raw transaction data
 - Directly contributed to securing a \$500,000 enterprise contract through technical leadership and client demonstration
- 2019 UCL-DiDi Master Research Program, Intern, DiDi AI Labs, Beijing, China
- Supervisor: Prof. Jieping Ye (VP of DiDi Chuxing, IEEE Fellow).
- Project: Leveraging Graph Neural Networks for User Travel Intent Prediction.
- Selected as one of only 2 projects (out of 30+ candidates) across the entire UCL cohort for the DiDi AI Labs research program.
- This work laid the foundation for applying GNNs to financial fraud detection at RaptorX.

Patents & Publications

Patents (inventor): 20 applications spanning graph-based fraud detection and LLM/agent systems, with counsel: 6 filed (US), 4 published (India), 10 in draft. Published (India) App Nos.: 202541024623, 202541045354, 202441079562, 202541032797.

Publications (first author; all under review):

- When the Tool Decides: LLM Agents Defer Blindly to GNN Tools (TMLR; arXiv:2606.14476)
- LLM Features Can Hurt GNNs: Concatenation Interference on Homophilous Graph Benchmarks (TMLR; arXiv:2606.17579)
- How Much Do Deep GNNs Actually Help? Decomposing Depth, Features, and Structure (NeurIPS 2026 E&D)

Technical Skills

- Agent / ML Agentic engineering, LLM APIs, function calling/tool use, guardrails, agent evaluation, PyTorch, PyG, GNNs, autoencoders, fraud-ring detection
- Data & MLOps Spark / PySpark, Kafka, ClickHouse, Redis, Parquet, Docker, Kubernetes, ArgoCD, CI/CD, FastAPI / gRPC, MLflow
- Cloud AWS (S3 / EC2 / ECR), cloud GPU workflows, Linux, Grafana, Prometheus, Loki

Education

- 2018–2019 University College London, London, UK.
MSc in Data Science and Machine Learning.
Thesis: Graph Convolutional Networks for user behavior prediction (jointly with DiDi AI Labs).
Achieved top 3% in group NLP research project among 300 students.
- 2015–2018 King's College London, London, UK.
BSc in Computer Science (Artificial Intelligence).